



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

22-08-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S & I(A)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	B	C	C	B	A
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	B,C	A,C,D	A,B,C,D	A,D	A,B,C,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.33	0.01 to 0.02	80.00	233.00	2.00	4.50

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	C	C	D	B	A
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C	B,C	A,C	A,B	A,C	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	374.30	729.00	5.00	0.00	2.50	4.40

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	B	B	C	B	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	C,D	A,B	A,C,D	A,B	B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	9.00	6.00	5.00	0.72 to 0.74	0.72 to 0.74	0.00

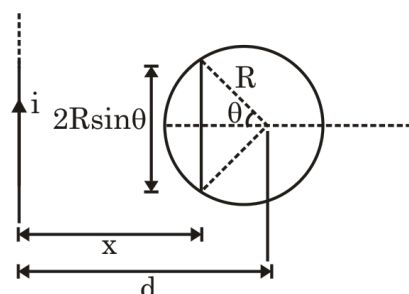
HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

To find mutual inductance, let's flow current in wire.



$$x = d - R \cos \theta$$

$$dx = +R \sin \theta d\theta$$

$$d\phi = BdA = \left(\frac{\mu_0 i}{2\pi x} \right) (2R \sin \theta) dx$$

$$M_i = \frac{\mu_0 i R}{\pi} \int_{\theta=0}^{\pi} \frac{\sin \theta (R \sin \theta) d\theta}{(d - R \cos \theta)}$$

$$M = \frac{\mu_0 R^2}{\pi} \int_0^{\pi} \frac{(1 - \cos \theta)(1 + \cos \theta)}{R(1 - \cos \theta)} d\theta$$

$$= \frac{\mu_0 R}{\pi} [\theta + \sin \theta]_0^{\pi} = \frac{\mu_0 R}{\pi} (\pi) = \mu_0 R$$

2. Ans (B)

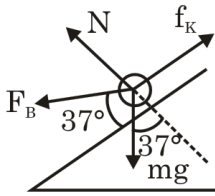
flux through coil will not change suddenly

$$\Rightarrow L \Delta I = -BA \cos \theta$$

$$\Delta I = -\frac{BA \cos \theta}{L} = -\frac{(2T)(0.2m)^2 \cos 37^\circ}{0.2h} = 0.32A$$

$$I_f = I_i + \Delta I = 1A - 0.32A = 0.68A$$

$$F_B = I \ell B \sin \theta = (0.68A)(0.2m)(2T) \sin 90^\circ = 0.272N$$



$$N = mg \cos 37^\circ - F_B \sin 37^\circ$$

$$= \frac{4}{5} - 0.272 \times \frac{3}{5} = \frac{3.484}{5} N$$

$$f_k = \mu N = \frac{3}{4} \times \frac{3.484}{5} = 0.5226 N$$

$$a = \frac{mg \sin 37^\circ + F_B \cos 37^\circ - f_k}{m}$$

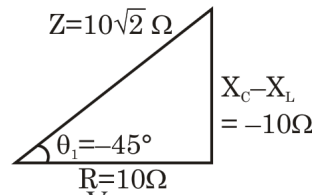
$$= \left[\frac{3}{5} + 0.272 \times \frac{4}{5} - 0.5226 \right] 10 \text{ m/s}^2$$

$$= 2.95 \text{ m/s}^2$$

3. Ans (C)

$$X_L = \omega L = 100 \times 0.1 \Omega = 10 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 2 \times 10^{-3}} \Omega = 5 \Omega$$

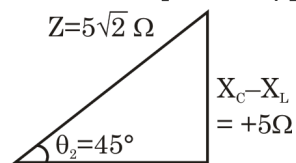


$$I_L = \frac{V_0}{Z} \sin(\omega t + \phi_1)$$

$$= \sqrt{2} \sin\left[100t - \frac{\pi}{4}\right]$$

$$V_L = I_{L_0} X_L \sin\left[\omega t + \phi_1 + \frac{\pi}{2}\right]$$

$$= 10\sqrt{2} \sin\left[100t + \frac{\pi}{4}\right]$$



$$I_C = \frac{V_0}{Z} \sin(\omega t + \phi_2) = 2\sqrt{2} \sin\left(100t + \frac{\pi}{4}\right)$$

$$I_0 = \sqrt{I_C^2 + I_L^2} = \sqrt{(\sqrt{2})^2 + (2\sqrt{2})^2} = \sqrt{10} \text{ A}$$

$$I_{\text{RMS}} = \frac{I_0}{\sqrt{2}} = \sqrt{5} \text{ A}$$

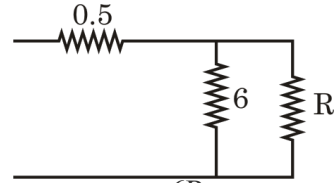
$$V_C = I_{C_0} X_{C_0} \sin\left(\omega t + \phi_2 - \frac{\pi}{2}\right)$$

$$= 10\sqrt{2} \sin\left(\omega t - \frac{\pi}{4}\right)$$

$$V_0 = \sqrt{V_{L_0}^2 + V_{C_0}^2} = 20 \text{ volt}$$

$$V_{\text{RMS}} = \frac{V_0}{\sqrt{2}} = 10\sqrt{2} \text{ volt}$$

4. Ans (C)



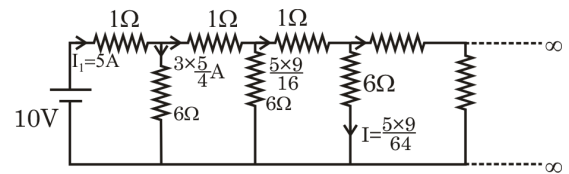
$$R = 0.5 + \frac{6R}{6+R}$$

$$R = \frac{3+6.5R}{6+R}$$

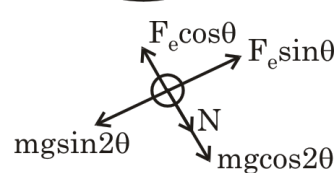
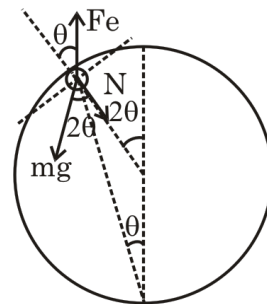
$$6R + R^2 = 3 + 6.5R$$

$$R^2 - 0.5R - 3 = 0$$

$$R = 2 \Omega$$



5. Ans (B)



for equilibrium t_0 be stable, net force must be towards mean position for small displacement

$$F_e \sin \theta \geq mg \sin 2\theta$$

$$\text{or } F_e \theta \approx mg(2\theta)$$

$$F_e \approx 2mg$$

$$\frac{Qq}{4\pi\epsilon_0 (2R)^2} \approx 2mg \Rightarrow q = \frac{32\pi\epsilon_0 mgR^2}{Q}$$

6. Ans (A)

$$E = -\frac{GM^2}{2R_0} + \frac{1}{2} M v_0^2 = -\frac{GM^2}{2r}$$

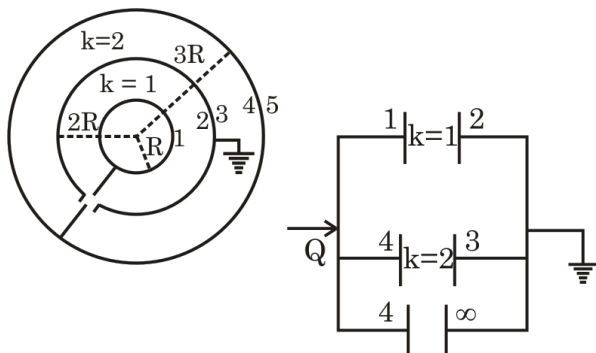
$$\frac{GM}{r} = \frac{GM}{R_0} - v_0^2 = \frac{GM - v_0^2 R}{R_0}$$

$$r = \frac{GMR_0}{GM - v_0^2 R}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B,C)



$$C_{12} = 8\pi\epsilon_0 R$$

$$C_{34} = 48\pi\epsilon_0 R$$

$$C_{400} = 12\pi\epsilon_0 R$$

$$C_{eq} = C_{12} + C_{34} + C_{400} = 68\pi\epsilon_0 R$$

$$q_{12} = \frac{QC_{12}}{C_{eq}} = \frac{8}{68}Q = \frac{2}{17}Q$$

$$q_{43} = \frac{QC_{34}}{C_{eq}} = Q \left(\frac{48}{68} \right) = \frac{12}{17}Q$$

$$\rightarrow Q_2 = -(q_{12} + q_{43}) = -\frac{14}{17}Q$$

8. Ans (A,C,D)

$$v_P = -v \frac{dy}{dx}$$

$$20\sqrt{3} = -v \tan 60^\circ$$

$v = -2\text{cm/s}$, here $-ve$ represent that wave is moving in $-ve$ x direction

$$A = 4\text{mm}, \lambda = (5.5 + 5)\text{cm} = 4\text{cm}$$

$$\text{at } t = 0, x = 0, y = 2\sqrt{2}\text{mm}$$

$$y = A \sin \left[\frac{2\pi}{\lambda} (v_t + x) + \phi \right]$$

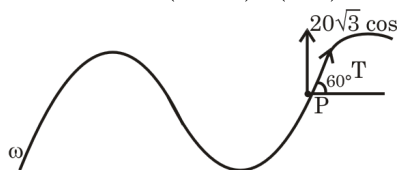
$$y = 4\text{mm} \sin \left[\frac{2\pi}{4\text{cm}} \left(20 \frac{\text{cm}}{\text{s}} t + x \right) + \frac{\pi}{4} \right]$$

$$= 4\text{mm} \sin \left[10\pi t + 50\pi x + \frac{\pi}{4} \right]$$

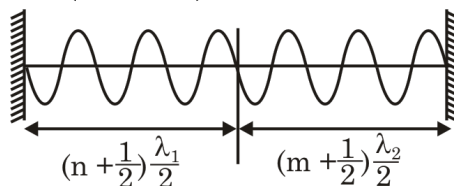
$$v = \sqrt{\frac{T}{\mu}} \Rightarrow T = \mu v^2 = (0.1)(0.2)^2 \text{N} = 4 \times 10^{-3} \text{N}$$

$$P = T v_P \cos \theta$$

$$= (4 \times 10^{-3}) \left(\frac{20\sqrt{3}}{100} \right) \left(\frac{\sqrt{3}}{2} \right) \omega = 12 \times 10^{-5} \omega$$



9. Ans (A,B,C,D)



$$\frac{v_1}{v_2} = \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{\mu_2}{\mu_1}} = \frac{2}{1}$$

$$\frac{\ell_1}{\ell_2} = \frac{(2n+1)\lambda_1}{(2m+1)\lambda_2} = \frac{(2n+1)}{(2m+1)} \frac{1}{2}$$

$$\frac{5m}{6m} = \frac{(2n+1)}{(2m+1)} \frac{1}{2}$$

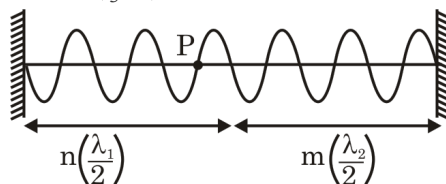
$$\frac{2n+1}{2m+1} = \frac{5}{3} \Rightarrow n = 2, m = 1$$

$$f_0 = \frac{v_1}{\lambda_1} = \frac{10\text{m/s}}{1\text{m}} = 10\text{Hz}$$

$$\text{for } 1^{\text{st}} \text{ overtone } \frac{2n+1}{2m+1} = \frac{15}{9}$$

$$\Rightarrow n = 7, m = 4$$

$$f_1 = \frac{10\text{m/s}}{\left(\frac{1}{3}\text{m}\right)} = 30\text{Hz}$$



$$\frac{\ell_1}{\ell_2} = \frac{n \left(\frac{\lambda_1}{2} \right)}{m \left(\frac{\lambda_2}{2} \right)}$$

$$\frac{5}{6} = \frac{n}{m} \times \frac{1}{2} \Rightarrow \frac{n}{m} = \frac{5}{3}, \frac{10}{6}, \dots$$

$$f_0 = \frac{v_1}{\lambda_1} = \frac{10\text{m/s}}{2\text{m}} = 5\text{Hz}$$

$$f_1 = \frac{v_1}{\lambda_1} = \frac{10\text{m/s}}{1\text{m}} = 10\text{Hz}$$

10. Ans (A,D)

$$\frac{\frac{4}{3}}{v} - \frac{1}{-60} = \frac{\frac{3}{2} - 1}{+20} + \frac{\frac{4}{3} - \frac{3}{2}}{-30}$$

$$\frac{4}{3v} = \frac{1}{40} + \frac{1}{180} - \frac{1}{60}$$

$$= \frac{9+2-6}{360} = \frac{5}{360} = \frac{1}{72}$$

$$v = +96\text{cm}$$

now if this image form at centre of curvature or pole of minor than final image will form on object itself

$$\Rightarrow x = v - 2f = 96 + 60 = 156\text{cm}$$

$$x = v = 156\text{cm}$$

11. Ans (A,B,C,D)

$$a_2 = r_2 \theta$$

$$a_1 = r_1 \theta$$

$$\frac{q_1}{q_2} = \frac{\lambda a_1}{\lambda a_2} = \frac{r_1}{r_2}$$

$$\frac{v_1}{v_2} = \frac{\frac{kq_1}{r_1}}{\frac{kq_2}{r_2}} = \frac{q_1}{q_2} \times \frac{r_2}{r_1} = 1$$

$$\frac{E_1}{E_2} = \frac{\frac{kq_1}{r_1^2}}{\frac{kq_2}{r_2^2}} = \frac{q_1}{q_2} \times \left(\frac{r_2}{r_1}\right)^2 = \frac{r_2}{r_1} > 1$$

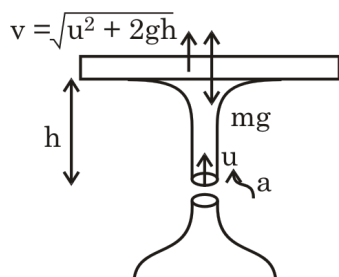
$$\Rightarrow E_1 > E_2$$

i.e. net field will be towards a_2

PART-1 : PHYSICS

SECTION-II

1. Ans (3.33)



$$F + h = \rho A v^2 = \rho a u \sqrt{u^2 - 2gh}$$

$$mg = \rho a u \sqrt{u^2 - 2gh} \dots\dots(1)$$

$$\rho a u \sqrt{u^2 - 2g(h + \Delta h)} - mg = ma$$

$$\rho a u \sqrt{u^2 - 2gh} \left\{ \sqrt{1 - \frac{2g\Delta h}{u^2 - 2gh}} \right\} - mg = ma$$

$$m(\omega) = 10^3 \times 15 \times 10^{-4} \times 5 \times 3$$

$$m = \frac{225}{100} = 2.25 \text{ kg}$$

$$\rho a u \sqrt{u^2 - 2gh} \left(1 - \frac{g\Delta h}{u^2 - 2gh} \right) - mg = ma$$

$$ma = - \frac{\rho a u g}{\sqrt{u^2 - 2gh}} \Delta h$$

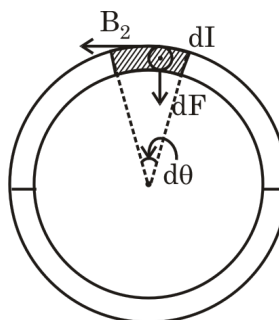
$$a = - \frac{\rho a u g}{m \sqrt{u^2 - 2gh}} \Delta h = -\omega^2 \Delta h$$

$$\Rightarrow \omega = \sqrt{\frac{\rho a u g}{m \sqrt{u^2 - 2gh}}} = \sqrt{\frac{10^3 \times 15 \times 10^{-4} \times 5 \times 10}{2.25 \sqrt{25 - 2(10)(0.8)}}$$

$$= \sqrt{\frac{15 \times 8}{18 \times 15 \times 3}} = \sqrt{\frac{100}{9}} = \frac{10}{3} \text{ rad/s}$$

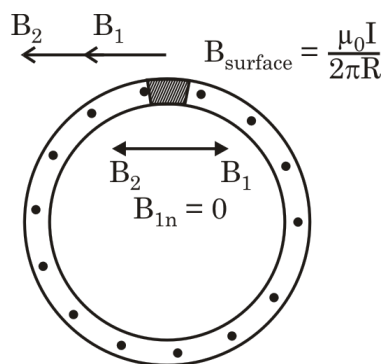
$$= 3.33 \text{ rad/s}$$

2. Ans (0.01 to 0.02)



$$dI = \frac{I}{2\pi R} (R d\theta) = \frac{I d\theta}{2\pi}$$

$$dF = (dI) B_2 \ell$$



$$B_2 - B_1 = 0$$

$$B_2 + B_1 = \frac{\mu_0 I}{2\pi R}$$

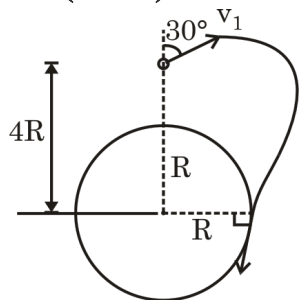
$$\Rightarrow B_1 = B_2 = \frac{\mu_0 I}{4\pi R} \dots(1)$$

$$dF = \left(\frac{I d\theta}{2\pi} \right) \left(\frac{\mu_0 I}{4\pi R} \right) \ell$$

$$\text{Pressure} = \frac{dF}{dA} = \frac{\frac{\mu_0 I^2}{8\pi^2 R} \ell d\theta}{(R d\theta) \ell} = \frac{\mu_0 I^2}{8\pi^2 R^2}$$

$$= \frac{4\pi \times 10^{-7} (10)^2}{8\pi^2 (10^{-2})^2} = \frac{1}{20\pi} \text{ Pa} \approx 0.0159 \text{ Pa}$$

3. Ans (80.00)



From conservation of angular momentum
 $mv_1(4R)\sin 30^\circ = mv_2 R \sin 90^\circ$

$$\frac{v_2}{v_1} = 2 \dots\dots\dots (1)$$

From conservation of energy

$$-\frac{GMm}{4R} + \frac{1}{2}mv_1^2 = -\frac{GMm}{R} + \frac{1}{2}mv_2^2$$

$$-\frac{GM}{4R} + \frac{v_1^2}{2} = -\frac{GM}{R} + \frac{v_2^2}{2}$$

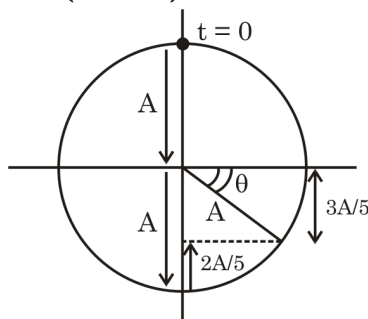
$$\frac{3v_1^2}{2} = \frac{3GM}{4R}$$

$$v_1 = \sqrt{\frac{GM}{2R}} = \sqrt{\frac{64 \times 10^6}{2}} = \frac{8 \times 10^3}{\sqrt{2}} \text{ m/s}$$

$$50\sqrt{2}x = 4\sqrt{2} \times 10^3$$

$$x = 80$$

4. Ans (233.00)



$$\sin \theta = \frac{3}{5}$$

$$\theta = \frac{37}{180} \pi$$

$$\text{Change in phase} = \pi + \frac{53}{180} \pi = \frac{233}{180} \pi$$

$$\text{time taken} = \frac{233}{360} T = 233 \text{ s}$$

5. Ans (2.00)

$$A > 2\theta_C$$

$$60^\circ > 2\theta_C$$

$$\text{or } \theta_C < 30^\circ$$

$$\sin \theta_C < \sin 30^\circ$$

$$\frac{1}{\mu} < \frac{1}{2}$$

$$\mu > 2$$

6. Ans (4.50)

At $t = 0$ current through inductor will be zero

$$\Rightarrow I_1 = \frac{10V}{5\Omega} = 2A$$

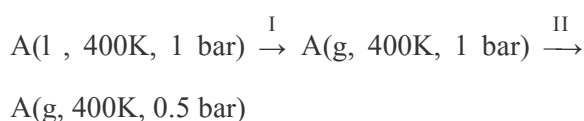
At $\tau \rightarrow \infty$, current through capacitor will be zero

$$\Rightarrow I_2 = \frac{10V}{4\Omega} = 2.5A$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (A,C)



$\Delta H_1 = +20 \text{ kcal}$ and $\Delta H_2 = 0$ (Enthalpy of ideal gas is independent of pressure)

$$\therefore \Delta H = \Delta H_1 + \Delta H_2 = +20 \text{ kcal}$$

$$\Delta S_1 = \frac{\Delta H_1}{T} = \frac{+20 \times 10^3 \text{ cal/mol}}{400K} = 50 \text{ cal/K-mol}$$

8. Ans (B,C)

Catalyst decrease the activation energy of reaction and changes the path. it does not changes the spontaneity at reaction.

For elementary reaction order and Molecularity are same

Rate constant depend on temperature

$$k = Ae^{-E_a/RT}$$

at low temperature rate is more temperature sensitive.

9. Ans (A,C)

$$\Delta H = 3 \times (-390) - 2(-820)$$

$$= 470 \text{ kJ/mol}$$

$$\Delta G < 0 \quad \Delta H - T \Delta S < 0$$

for spontaneous reaction $\rightarrow T \Delta S > \Delta H$

$$(\because \Delta S > 0)$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (374.30)

Given : molality = 1 molal ; $K_b = 0.5 \text{ K.kg/mol}$

$$\% \alpha = 80\%, \alpha = 0.8$$



$$1 - \alpha \quad \alpha \quad 2\alpha$$

$$\Rightarrow i = 1 - \alpha + \alpha + 2\alpha$$

$$i = 1 + 2\alpha = 1 + 2(0.8)$$

$$i = 2.6$$

Elevation in Boiling point, $\Delta T_b = iK_b m = 2.6 \times 0.5 \times 1$

$$\Delta T_b = 1.3$$

$$\text{Boiling point of solution} = T_b^0 + \Delta T_b = 373 + 1.3$$

$$= \boxed{374.3 \text{ K}}$$

2. Ans (729.00)



$$K_p = \frac{P_{O_2}^2 \times P_{SO_2}^4}{P_{SO_3}^4}$$

$$= 3^2 \times \left(\frac{6}{2}\right)^4 = 729 \text{ atm}^2$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

$$AB = \begin{pmatrix} ax^2 & bx^2 & cx^2 \\ a & b & c \\ 2ax & 2bx & 2cx \end{pmatrix}$$

$$\text{tr}(AB) = \text{tr}(C)$$

$$ax^2 + b + 2cx = 3x^2 + 4 + 6x$$

$$\Rightarrow a = 3 \quad b = 4 \quad c = 3$$

$$I = \int_0^{\infty} \frac{\ln x}{3x^2 + 4x + 3} dx$$

$$\text{Put } x = \frac{1}{y}$$

$$I = - \int_0^{\infty} \frac{\ln y}{3y^2 + 4y + 3} dy$$

$$\text{Replace } y \rightarrow x$$

$$I = - \int_0^{\infty} \frac{\ln x}{3x^2 + 4x + 3} dx$$

$$I = -I$$

$$HS-6/8 \Rightarrow I = 0$$

2. Ans (B)

$$3y^2 x^9 dy + y^3 dy = x^3 dx - 9x^8 y^3 dx$$

$$(3y^2 dy)x^9 + y^3(9x^8 dx) = x^3 dx - y^3 dy$$

$$x^9 d(y^3) + y^3 d(x^9) = x^3 dx - y^3 dy$$

$$d(x^9 y^3) = x^3 dx - y^3 dy$$

$$4x^9 y^3 = x^4 - y^4 + c$$

It passes through (0, 0) $\Rightarrow c = 0$

$$4(x^3 y)^3 + y^4 - x^4 = 0$$

3. Ans (B)

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1 - \lambda & 0 & 0 \\ 0 & 1 - \lambda & 2 \\ 0 & -2 & 3 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (1 - \lambda) [(1 - \lambda)(3 - \lambda) + 4] = 0$$

$$\Rightarrow \lambda^3 - 5\lambda^2 + 11\lambda - 7 = 0$$

$$\text{we have, } A^3 - 5A^2 + 11A - 7I = 0$$

$$A^7 - 4A^6 + 6A^5 = (A^3 - 5A^2 + 11A - 7I)$$

$$(A^4 + A^3 - 4A - 13I) + (-21A^2 + 115A - 91I)$$

$$\Rightarrow A^7 - 4A^6 + 6A^5 = -21A^2 + 115A - 91I$$

4. Ans (C)

$$\frac{\sqrt{x^2 - y^2}}{\sqrt{1 + x^2 - y^2}} (x dx - y dy) = x dy - y dx$$

$$\frac{(x^2 - y^2) d(x^2 - y^2)}{2\sqrt{1 + x^2 - y^2} \sqrt{x^2 - y^2}} = x^2 \left(\frac{x dy - y dx}{x^2} \right)$$

$$\int \frac{d(x^2 - y^2)}{2\sqrt{x^2 - y^2} \sqrt{1 + x^2 - y^2}} = \int \frac{x^2}{x^2 - y^2} d\left(\frac{y}{x}\right)$$

$$x^2 - y^2 = t^2$$

$$\int \frac{2t dt}{2t\sqrt{1 + t^2}} = \int \frac{d\left(\frac{y}{x}\right)}{1 - \frac{y^2}{x^2}}$$

$$\ln |t + \sqrt{1 + t^2}| = \frac{1}{2} \ln \left| \frac{1 + \frac{y}{x}}{1 - \frac{y}{x}} \right| + \ln c$$

$$\ln |t + \sqrt{1 + t^2}| = \ln c \sqrt{\frac{x + y}{x - y}}$$

$$\sqrt{x^2 - y^2} + \sqrt{1 + x^2 - y^2} = c \frac{(x + y)}{\sqrt{x^2 - y^2}}$$

$$f(x, y) = \sqrt{x^2 - y^2}$$

5. **Ans (B)**

$$\lim_{n \rightarrow \infty} \left\{ \sqrt{n^2 + n + 1} \right\} = \lim_{n \rightarrow \infty} \left\{ n \sqrt{1 + \frac{1}{n} + \frac{1}{n^2}} \right\}$$

$$= \lim_{n \rightarrow \infty} \left\{ n \left(1 + \frac{1}{2} \left(\frac{1}{n} + \frac{1}{n^2} \right) + \frac{\frac{1}{2} \left(-\frac{1}{2} \right)}{2!} \left(\frac{1}{n} + \frac{1}{n^2} \right)^2 + \dots \right) \right\}$$

$$= \lim_{n \rightarrow \infty} \left\{ n + \frac{1}{2} + \frac{1}{2n} - \frac{1}{8} \frac{1}{n} \left(1 + \frac{1}{n} \right)^2 \dots \right\} = \frac{1}{2}$$

6. **Ans (D)**

$$\text{Number of ways} = {}^5C_4 2^4 = 80$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A,C,D)**

$$\text{Let } y = \frac{x^3 + 1}{2} = \sqrt[3]{2x - 1}$$

$$2y - 1 = x^3 \text{ and } 2x - 1 = y^3$$

$$\text{subtract to get : } (y - x) [x^2 + y^2 + xy + 2] = 0$$

$$x = y \text{ or } x^2 + y^2 + xy + 2 = 0$$

$$\Rightarrow \left(x + \frac{y}{2} \right)^2 + \frac{3y^2}{4} + 2 = 0 \text{ (Not possible)}$$

$x = y$ is the only possibility

$$\frac{x^3 + 1}{2} = x \Rightarrow x^3 - 2x + 1 = 0$$

$$(x - 1)(x^2 + x - 1) = 0$$

$$x = 1, \frac{-1 \pm \sqrt{5}}{2}$$

$$a = 1, b = -1, c = 5, d = 2$$

8. **Ans (C,D)**

$$f(x) = \int_0^\pi \cos t (\cos x \cos t - \sin x \sin t) dt$$

$$f(x) = \int_0^\pi \cos x \cos^2 t dt - \sin x \int_0^\pi \sin t \cos t dt$$

$$f(x) = \frac{\pi}{2} \cos x - 0$$

$$f'(x) = 0 \text{ occurs at } x = \pi \text{ and min. value of } f(x)$$

$$\text{is } -\frac{\pi}{2}$$

9. **Ans (A,B)**

$$g(x + y) = e^y g(x) + e^x g(y)$$

$$\Rightarrow g(0) = g(0) + g(0) \Rightarrow g(0) = 0$$

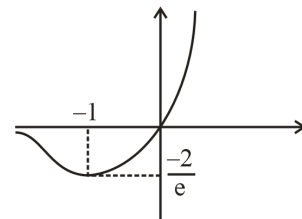
$$g'(x + y) = e^y g'(x) + e^x g'(y)$$

replace x by 0 and y by x

$$g'(x) = e^x g'(0) + e^0 g(x)$$

$$g'(x) - g(x) = 2e^x \Rightarrow g(x) = (2x + c)e^x$$

$$g(0) = c = 0$$



$$\text{Hence, } g(x) = 2xe^x$$

$$g'(x) = 2(x + 1)e^x$$

$$g''(x) = 2(x + 2)e^x$$

$$g(x) \text{ is } \uparrow \forall x \in [-1, \infty)$$

$$R_{g(x)} \leftarrow \left[\frac{-2}{e}, \infty \right)$$

$$\lim_{x \rightarrow 0} \frac{g(x)}{x} = 2$$

10. **Ans (A,C,D)**

$$\left. \begin{aligned} \frac{dy}{dx} + Py = Q \\ \frac{dy_1}{dx} + Py_1 = Q \end{aligned} \right\} \frac{d(y - y_1)}{dx} + P(y - y_1) = 0$$

$$\left. \begin{aligned} \frac{dy_2}{dx} + Py_2 = Q \\ \frac{d(y - y_1)}{dx} + P(y - y_1) = 0 \end{aligned} \right\} \frac{d(y - y_1)}{dx} = -P dx \Rightarrow y - y_1 = \lambda_1 e^{-\int P dx}$$

$$\|y - y_1 = \lambda_2 e^{-\int P dx}$$

$$\text{Hence } \frac{y - y_1}{y_2 - y_1} = \frac{\lambda_1}{\lambda_2} = c \text{ or } y = y_1 + c(y_2 - y_1)$$

If $\alpha y_1 + \beta y_2$ is a solution,

$$\text{hence } \frac{d}{dx}(\alpha y_1 + \beta y_2) + P(\alpha y_1 + \beta y_2) = 0$$

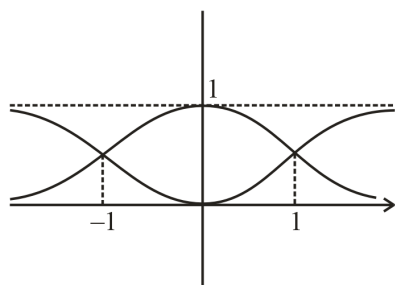
$$\alpha \left(\frac{dy_1}{dx} + P y_1 \right) + \beta \left(\frac{dy_2}{dx} + P y_2 \right) = 0$$

$$\alpha Q + \beta Q = 0 \Rightarrow \alpha + \beta = 0$$

$$\|y - \alpha y_1 - \beta y_2 \text{ is a solution if } \alpha - \beta = 1$$

11. Ans (A,B)

$$y = \frac{1}{1+x^2}, y = 1 - \frac{1}{1+x^2}$$



$$\text{Area} = \int_{-1}^1 \left(\frac{1}{1+x^2} - \left(1 - \frac{1}{1+x^2} \right) \right) dx$$

$$= 2 \int_0^1 \left(\frac{2}{1+x^2} - 1 \right) dx$$

$$4 \tan^{-1} x \Big|_0^1 - 2 = \pi - 2$$

12. Ans (B,D)

(A) is incorrect

(B) Number of ways to select p items from n-1 items = ${}^{n-1}C_p$

(C) Use integral solution method to get ${}^{n-1}C_{p-1}$

$$(D) |x|x|x|x| \dots |x| \Rightarrow {}^{n-1}C_p \frac{\frac{n-2}{n-2}}{\frac{p}{p}} = {}^{n-1}C_p$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (9.00)

$$(x^2 - 3x - 4)(x^2 + 2x - 3) < 0$$

$$(x-4)(x+1)(x+3)(x-1) < 0$$

$$\Rightarrow x \in [-3, -1] \cup [1, 4]$$

2. Ans (6.00)

$$\text{Let } x = (\alpha \gamma ABC^2 D)^3$$

$$|x| = |\alpha \gamma ABC^2 D|^3$$

$$\therefore \text{order of } ABC^2 D \text{ is } 2 \times 2$$

$$\therefore |x| = (\alpha^2 \gamma^2)^3 |ABC^2 D|^3$$

$$\therefore k = (\alpha \gamma)^6 \Rightarrow \lambda = 6$$

3. Ans (5.00)

$$A = \frac{d(\tan^{-1} \frac{x}{2})}{d(\tan^{-1} \frac{x}{2})} \Big|_{x=0} = 1$$

$$B = \frac{d(\sin^{-1} x)}{d(\cos^{-1} x)} \Big|_{x=\frac{1}{2}} = -1$$

$$2A + 3B^2 = 5$$

4. Ans (0.72 to 0.74)

$$I_1 = \int_0^1 \frac{dx}{e^x (1+x)},$$

$$I_2 = \int_0^{\pi/4} \frac{e^{\tan^2 \theta}}{(2 - \tan^2 \theta)} \tan \theta \sec^2 \theta d\theta$$

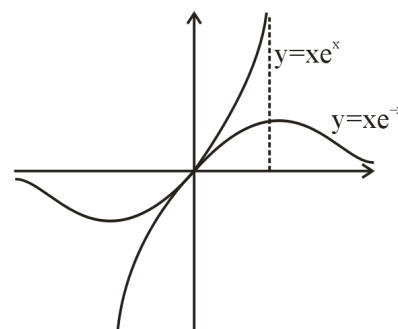
$$\text{Let } \tan^2 \theta = t$$

$$I_2 = \int_0^1 \frac{e^t dt}{2(2-t)} \quad \text{Let } 1-t = x$$

$$\Rightarrow I_2 = \int_1^0 \frac{e^{1-x} (-dx)}{2(1+x)}$$

$$I_2 = \int_0^1 \frac{e}{2} \frac{dx}{e^x (1+x)} = \frac{e}{2} \quad I_1 = \frac{I_2}{I_2} = \frac{2}{e}$$

5. Ans (0.72 to 0.74)



$$xe^x = xe^{-x} \Rightarrow x = 0$$

$$\text{Area} = \int_0^1 xe^x - xe^{-x} dx = \frac{2}{e} \text{ sq. units}$$

6. Ans (0.00)

$$|B| = \begin{vmatrix} a_{11} & 3^{-1}a_{12} & 3^{-2}a_{13} \\ 3a_{21} & a_{22} & 3^{-1}a_{23} \\ 3^2a_{31} & 3a_{32} & a_{33} \end{vmatrix} = |A|$$

$$\text{Similarly, } |C| = |B|$$

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S & I(A)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	B,C,D	A,B,D	A,B	A,B	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.37	1.25	8.46	9.00	10.00	0.45
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	5	1	3	2	1

PART-2 : CHEMISTRY

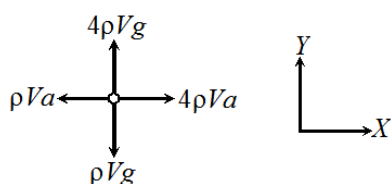
SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,B,D	A,C	D	B,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	500.00	0.50	7.00	4.00	3.00	4.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	7	4	6	1	3

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C	A,B,C,D	A,B	B,C,D	A,B,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	6.00	9.00	1.00	439.00	3.00	4.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	0	5	1	3	3	4

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (B,C)



F.B.D. of ball in frame of container

$$a_y = \frac{4\rho V g - \rho V g}{\rho V} = 3g$$

$$\frac{L}{2} = \frac{1}{2} 3gt^2 \Rightarrow t = \sqrt{\frac{L}{3g}}$$

$$a_x = \frac{4\rho V a - \rho V a}{\rho V} = 3a$$

$$L = \frac{1}{2} 3at^2 \Rightarrow t = \sqrt{\frac{2L}{3a}}$$

Ball to collide at point Q,

$$t = \sqrt{\frac{2L}{3a}} = \sqrt{\frac{L}{3g}} \Rightarrow a = 2g \therefore (B, C)$$

2. Ans (B,C,D)

If equation of a wave pulse is $y = f(ax \pm bt)$ The speed of wave is $\frac{b}{a}$ in negative x directionfor $y = f(ax + bt)$ and positive x direction for $y = f(ax - bt)$.Comparing this from given equation we can find the speed of wave is $5/4 = 1.25$ m/s and it is travelling in negative x-direction.Since $y(x) = y(-x)$ at $t = 0$ $\therefore$ pulse is symmetric, also maximum value of

$$Y \text{ is } \frac{0.8}{5} = 0.16 \text{ m}$$

 $\therefore$ (B), (C) and (D)

3. **Ans (A,B,D)**

Gravitational field intensity inside the spherical shell is zero.

Gravitational force is always directed towards the source mass.

Force on P act only due to m_1 because point P is outside the spherical shell of mass m_1 .

So choice (A), (B) and (D) are correct.

$\therefore$ (A), (B) and (D)

4. **Ans (A,B)**

Current in 3Ω resistance is 3 A

Power dissipated in $3\Omega = i^2 R = 27$ Watt

$V_C - 5 \times 1 + 12 - 8 \times 2 - 3 - 4 \times 2 = V_B$

$V_C - V_B = 20$ Volt

$V_C - V_D = 7$ Volt

Power dissipated in $6\Omega = 216$ Watt

$\therefore$ (A) and (B)

5. **Ans (A,B)**

q_1 and q_2 both are negative and

$$\frac{kq_1}{\left(\frac{2a}{3}\right)^2} = \frac{kq_2}{\left(\frac{a}{3}\right)^2} \Rightarrow \frac{q_1}{q_2} = 4$$

$\therefore$ (A) and (B)

6. **Ans (B,C)**

$$\left(P_a + \frac{2T}{r}\right) + \rho_1 gh = \rho_2 gh + P_a$$

$$\text{or } T = \frac{r}{2}(\rho_2 - \rho_1) gh$$

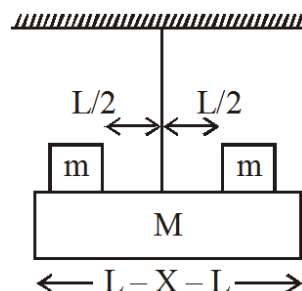
If surface tension is zero, height of liquid column inside the capillary will drop.

Ans. (B,C)

PART-1 : PHYSICS

SECTION-II

1. **Ans (0.37)**



$$f_1 = \frac{1}{2\pi} \sqrt{\frac{C}{I}} \dots(i)$$

$$= \frac{1}{2} \sqrt{\frac{3C}{ML^2}}$$

$$f_2 = \frac{1}{2\pi} \sqrt{\frac{C}{L^2 \left(\frac{M}{3} + \frac{m}{2}\right)}} \dots(ii)$$

As frequency reduces by 80%

$$\therefore f_2 = 0.8f_1 = \frac{f_2}{f_1} = 0.8 \dots(iii)$$

Solving equations (i), (ii) & (iii)

$$\text{Ratio } \frac{m}{M} = 0.37$$

Ans. 0.37

2. **Ans (1.25)**

The resistance of the loops,

$$R_1 = 2\pi r_1 \times 10 = 2\pi \times 0.1 \times 10$$

$$= 6.28 \Omega$$

$$\text{and } R_2 = 2\pi r_2 = 2\pi \times 1 \times 10$$

$$= 62.8 \Omega.$$

Flux in the smaller loop, $\phi = B_2 A_1$

$$= \frac{\mu_0 i_2}{2r_2} \pi r_1^2$$

$$= \frac{\mu_0 \left[\frac{V}{R_2}\right] \pi r_1^2}{2r_2}$$

$$= \frac{\mu_0 \frac{[4+2.5t]}{R_2} \pi r_1^2}{2r_2}$$

$$\text{The induced current, } i_1 = \frac{e}{R_1} = \frac{\left[\frac{d\phi}{dt}\right]}{R_1}$$

After substituting the value and simplifying we get

$$i = 1.25 \times 10^{-10} \text{ A.}$$

3. **Ans (8.46)**

Capacitance of a capacitor with a dielectric of dielectric constant k is given by

$$C = \frac{k\epsilon_0 A}{d}$$

$$\therefore E = \frac{V}{d} \quad \therefore C = \frac{k\epsilon_0 AE}{V}$$

$$15 \times 10^{-12} = \frac{k \times 8.86 \times 10^{-12} \times 10^{-4} \times 10^6}{500}$$

$$k = 8.46$$

4. **Ans (9.00)**

$$\frac{\rho gh}{2} \times hL = TL - TL \sin 30^\circ$$

$$\Rightarrow h = \sqrt{\frac{T}{\rho g}} = 9\text{mm}$$

Answer : 9.00

5. **Ans (10.00)**

$$q[v_x \hat{i} + v_y \hat{j}] \times B_0 \hat{k} - \alpha[v_x \hat{i} + v_y \hat{j}] = m \frac{d\vec{v}}{dt} \dots (i)$$

$$\therefore a_y = -\frac{1}{m}[qv_x B_0 + \alpha v_y] \text{ and}$$

$$a_x = \frac{1}{m}[qB_0 v_y - \alpha v_x] \dots (iii)$$

$$\therefore \text{At } t = 0, v_y = v_0 \text{ and } v_x = 0 \text{ and } t = t, v_x = 0, v_y = 0$$

$$\therefore v_0 = \frac{1}{m} \left[qB_0 x + \frac{\alpha^2 x}{qB_0} \right]$$

$$\Rightarrow x = \frac{qB_0 m v_0}{\alpha^2 + q^2 B^2} = 10$$

Answer. 10.00

6. **Ans (0.45)**

Here $Z = x + jy + \omega Lj = x + j(y + \omega L)$ for power factor to be one $y + \omega L = 0 \Rightarrow y = -10$

$$I = \frac{v_0}{x}, x = \frac{v_0}{I} = \frac{25}{5} = 5, \text{ Impedance of box} = 5 - 10j,$$

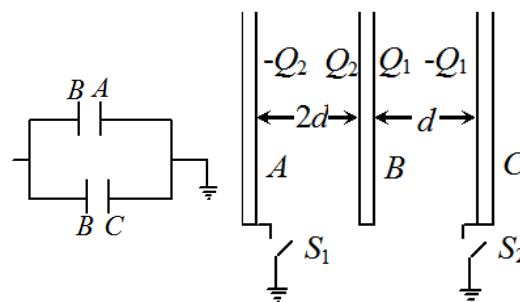
$$\cos \phi = \frac{5}{\sqrt{5^2 + 10^2}} = \frac{1}{\sqrt{5}} = 0.45$$

Answer. 0.45

PART-1 : PHYSICS

SECTION-III

1. **Ans (1)**



$$Q_1 + Q_2 = Q \dots (i)$$

$$\frac{Q_2}{C_2} = \frac{Q_1}{C_1} \quad \left(C_2 = \frac{A\epsilon_0}{2d}, C_1 = \frac{A\epsilon_0}{d} \right)$$

$$2Q_2 = Q_1 \dots (ii)$$

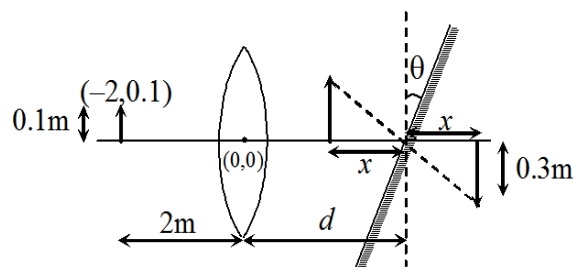
On solving (i) and (ii) we get

$$Q_1 = \frac{2Q}{3}, Q_2 = \frac{Q}{3}$$

$$\text{Charge on A} = -\frac{Q}{3} = +1\mu\text{C}$$

Answer. 1

2. **Ans (5)**



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-2} = \frac{1}{1.5}; \quad v = 6\text{m}$$

$$m = \frac{v}{u} = -3$$

$$x = 6 - d; \quad \tan \theta = \frac{0.3}{x}$$

$$\Rightarrow d = 5\text{m}.$$

Answer. 5

3. **Ans (1)**

When bar attains terminal velocity $BIL = mg$

$$I = \frac{0.2 \times 9.8}{0.6 \times 1} = \frac{9.8}{3},$$

e is emf induced in rod and $eI = P_1 + P_2$

$$e = \frac{0.76 + 1.2}{9.8/3} = \frac{1.96 \times 3}{9.8} = 0.6 \text{ V and}$$

$$e = Bv_T \ell$$

$$v_T = \frac{e}{B} = 1 \text{ m/s}$$

Answer: 1

4. **Ans (3)**

Initially the charge on either capacitor i.e., q_A or q_B is CV coulomb.

When dielectric is introduced, the new capacitance of either capacitor

$$C_1 = KC = 3C$$

After the opening of switch S , the potential across capacitor A is V volts.

Let the potential across capacitor B is V_1

$$\therefore q_B = CV = C_1 V_1 \text{ or } CV = 3 CV_1$$

$$\therefore V_1 = V/3 \text{ volts}$$

$$\text{Initial energy of capacitor } A = \frac{1}{2} CV^2$$

$$\text{energy of capacitor } B = \frac{1}{2} CV^2$$

$$\therefore \text{total energy } E_i = \frac{1}{2} CV^2 + \frac{1}{2} CV^2 = CV^2$$

Final energy of capacitor

$$A = \frac{1}{2} \times (3C) V^2 = \frac{3}{2} CV^2$$

Final energy of capacitor

$$B = \frac{1}{2} \times (3C) \left(\frac{V}{3}\right)^2 = \frac{CV^2}{6}$$

$\therefore$ Total final energy

$$E_f = \frac{3}{2} CV^2 + \frac{CV^2}{6} = \frac{5}{3} CV^2$$

$$\therefore \frac{E_i}{E_f} = \frac{CV^2}{(5/3) CV^2} = \frac{3}{5}, x = 3$$

Answer: 3

5. **Ans (2)**

Minimum kinetic energy needed to escape the particle

$$= \frac{1}{2} mv_e^2 = \frac{1}{2} m \frac{2GM}{R} = \frac{GMm}{R}$$

Here, M = mass of earth.

m = mass of particle

$$\text{Kinetic energy given to the particle} = \frac{GMm}{3R}$$

Let the maximum height attained by the particle be h .

From conservation of mechanical energy

$$\frac{GMm}{3R} + \left(-\frac{GMm}{R}\right) = -\frac{GMm}{R+h}$$

$$\Rightarrow R+h = \frac{3R}{2}$$

$$\Rightarrow h = \frac{R}{2} = \frac{6400}{2} = 3200 \text{ km}$$

$$\Rightarrow N = 2$$

Answer: 2

6. **Ans (1)**

By work energy theorem

$$mgz + BIlz - \int 2k(x+z)dz = 0$$

$$2mgz = 2k \left[\int_0^z x dz + \int_0^z z dz \right] \text{ (where } x \text{ is elongation in the equilibrium position)}$$

$$2mg = mg + kz$$

$$z = \frac{mg}{k} = \frac{BI\ell}{k}$$

$$\Rightarrow N = 1$$

Answer: 1

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (500.00)**

$$\Delta H^\circ_{\text{reaction}} = -375 - (-290) = -85 \text{ kJ/mole}$$

$$\Delta S^\circ = 360 - 220 - 310 = -170 \text{ J/mole/K}$$

For equilibrium $\Delta G^\circ = 0$

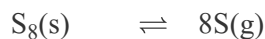
$$\Delta H^\circ = T\Delta S^\circ$$

$$T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{-85}{-170} \times 1000 = 500 \text{ K}$$

2. **Ans (0.50)**

$$\text{moles of } S_8 = \frac{2.56}{256} = 0.01$$

$$\text{moles of sulphur} = \frac{\frac{960}{1000} \times 1}{0.08 \times 300} = 0.04$$



$$0.01(1-\alpha) \quad 8 \times 0.01 \times \alpha$$

$$8 \times 0.01 \alpha = 0.04$$

$$\alpha = 0.5$$

4. **Ans (4.00)**

I, III, IV, VIII

5. **Ans (3.00)**

II, IV, V

6. **Ans (4.00)**

$$r = K[A]^n$$

$$r = K[A]$$

$$2 \times 10^{-4} = K[A]$$

$$K = 3 \times 10^{-3} \text{ min}^{-1}$$

$$K = [A] = \frac{3 \times 10^{-3}}{60} \text{ s}^{-1}$$

$$[A] = \frac{2 \times 10^{-4}}{K} = \frac{2 \times 10^{-4} \times 60}{3 \times 10^{-3}}$$

$$= 4$$

PART-2 : CHEMISTRY

SECTION-III

1. **Ans (1)**

$$\Delta U = 0$$

$$W_{\text{total}} = -q_{\text{total}}$$

$$80 + (-40) + W_{\text{CA}} = 20 + 40 - 30$$

$$W_{\text{CA}} = -10 \text{ J}$$

2. **Ans (7)**

$$K_I = \frac{\ln 2}{t_{1/2}} = \frac{0.7}{t_{1/2}}$$

$$K_{II} = \frac{0.1}{2 \times t_{1/2}}$$

$$\frac{K_I}{K_{II}} = 14 = X$$

3. **Ans (4)**

If KF crystallizes in NaCl type structure then K^+ is present in O.V. i.e. at edge centres and body center

$$2(r_{F^-} + r_{K^+}) = a =$$

$$2 \left(\frac{214}{\sqrt{2}} + \frac{186}{\sqrt{2}} \right) = 400\sqrt{2} \text{ pm}$$

shortest distance between K^- & K^+ ion

$$= \frac{a}{\sqrt{2}} = \frac{400\sqrt{2}}{\sqrt{2}} = 400 \text{ pm} = 4.00 \text{ \AA}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A,C)**

$$(A) f(x) = \ln(\tan \pi[x] + |x^2 + 2x - 3|)$$

$$\because [x] \in \mathbb{I} \Rightarrow \tan \pi[x] = 0,$$

$$\text{and } |x^2 + 2x - 3| = |(x+1)^2 - 2| \in [0, \infty)$$

So, $f(x) \in \mathbb{R} \Rightarrow f(x)$ is surjective

$$(B) g(x) = \frac{x^2 + 2x - 3}{x - 1}, x \neq 1$$

$$g(x) = \frac{(x-1)(x+3)}{(x-1)}, x \neq 1$$

$$g(x) = x + 3 \therefore g(x) \neq 4 \quad (\because x \neq 1)$$

So, range of $g(x)$ is $\mathbb{R} - \{4\}$

$\Rightarrow g(x)$ is not surjective

$$(C) h(x) = \ln \left(\frac{1-x}{1+x} \right), \frac{1-x}{1+x} > 0$$

$$\begin{array}{c} - \quad \quad + \quad \quad - \\ \hline \quad -1 \quad \quad 1 \end{array} \Rightarrow D_h = (-1, 1)$$

$$\because \frac{1-x}{1+x} \text{ take all value between } (0, \infty)$$

So, Range of $h(x) = \mathbb{R}$

$\Rightarrow h(x)$ is surjective

$$(D) k(x) = \sqrt{[x] + [-x] + 1} + \sqrt{\{x\} + \{-x\} + 1}$$

Domain of $k(x)$ is $\mathbb{R}$

$$x \notin \mathbb{I} \Rightarrow [x] + [-x] = -1 \text{ and } \{x\} + \{-x\} = 1$$

$$\Rightarrow k(x) = \sqrt{2}$$

$$x \in \mathbb{I} \Rightarrow [x] + [-x] = 0 \text{ and } \{x\} + \{-x\} = 0$$

$$\Rightarrow k(x) = 2$$

So, Range of $k(x) = \{\sqrt{2}, 2\}$

So, $k(x)$ is not surjective]

2. **Ans (A,B,C,D)**

Clearly $Q_3 = 0$ and $Q_4 = 2$ (2413,3142)

$Q_5 = 2[\text{start with } S_1] + 2[\text{start with } S_2] +$
[start with S_3]

$$= 2 \times 2 + 2 \times 3 + 4 = 14$$

In Q_{20} , S_{21} can be placed in 19 ways. Also there will be other ways like when S_{21} comes between S_1 and S_2 .

$$\therefore 19.Q_{20} < Q_{21}$$

3. **Ans (A,B)**

$$\text{We have } f(x) = e^x + \int_0^1 e^x f(t) dt = e^x + ke^x,$$

$$\text{where } k = \int_0^1 f(t) dt$$

$$\therefore k = \int_0^1 (e^t + ke^t) dt = e + ke - 1 - k$$

$$\text{So, } k = \frac{e-1}{2-e}$$

$$\therefore f(x) = e^x \left(1 + \frac{e-1}{2-e} \right) = \frac{e^x}{2-e}$$

$$\text{Obviously, } f(0) = \frac{1}{2-e} < 0$$

$$\text{Also } f'(x) = \frac{e^x}{2-e} < 0 \quad \forall x \in \mathbb{R}$$

Hence, $f(x)$ is a decreasing function on $\mathbb{R}$.

$$\text{Also } \int_0^1 f(x) dx = \int_0^1 \frac{e^x}{2-e} dx = \left[\frac{e^x}{2-e} \right]_0^1 =$$

$$\frac{e-1}{2-e} < 0]$$

4. **Ans (B,C,D)**

$$\text{Put } x = t^6$$

$$\therefore I = \int \frac{t^6 + t^4 + t}{t(1+t^2)} dt = 6 \int t^3 dt + \int \frac{dt}{1+t^2}$$

$$= \frac{3}{2}x^{\frac{2}{3}} + 6 \tan^{-1} \left(x^{\frac{1}{6}} \right) + \lambda$$

$$\therefore \alpha = \frac{3}{2}, \beta = \frac{2}{3}, \gamma = 6, \delta = \frac{1}{6}.$$

Sum of all coefficients of even power of x

$$= \frac{(1+6)^6 + (1-6)^6}{2}.]$$

5. **Ans (A,B,D)**

$$I = \int_{-1}^0 \left(\sqrt[3]{x^4 - 2x^2} - \sqrt{1 - \sqrt{x^3 + 1}} \right) dx$$

$$= - \int_{-1}^0 \left(\sqrt{1 - \sqrt{x^3 + 1}} - \sqrt[3]{x^4 - 2x^2} \right) dx$$

$$\text{Let } f(x) = \sqrt{1 - \sqrt{x^3 + 1}}$$

Hence, $g(y) = \sqrt[3]{y^4 - 2y^2}$ which is the inverse of $f(x)$.

$$\text{Now, } I = - \left(\int_{-1}^0 f(x) dx - \int_{-1}^0 g(y) dy \right)$$

$$= - \left(\int_{-1}^0 f(x) dx + \int_1^0 g(y) dy \right) \text{ On putting } y$$

$$= -t$$

$$\text{Hence, } I = -(1) = -1. \text{ Ans.}$$

6. **Ans (B,C)**

$$F(x) =$$

$$\int e^x \left(\int_0^x x \sin x^2 dx + \int_0^x (\sin x^2 + 2x^2 \cos x^2) dx \right) dx$$

$$= \int e^x \left(\int_0^x x \sin x^2 dx + (x \sin x^2) \right) dx$$

$$= \int e^x \left(\underbrace{\int_0^x x \sin x^2 dx}_{f(x)} + \underbrace{x \sin x^2}_{f'(x)} \right) dx$$

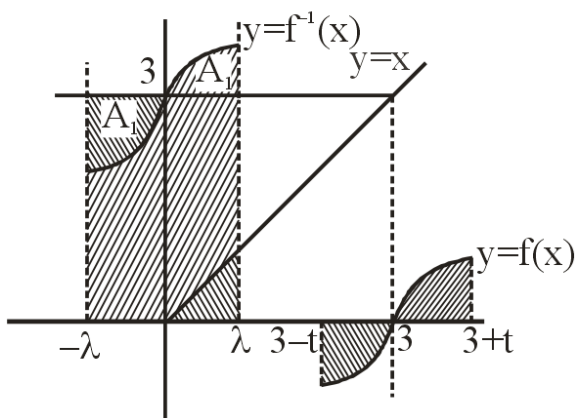
$$= e^x \int_0^x x \sin x^2 dx + c_1$$

$$= -\frac{1}{2} e^x \cos x^2 + c_1$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (6.00)



Required area = $2\lambda \times 3 = 6\lambda$

2. Ans (9.00)

$$f'(x) = \phi(x) \text{ and } \phi'(x) = f(x) \forall x \in \mathbb{R}$$

$$\text{Let } h(x) = f^2(x) - \phi^2(x)$$

$$h'(x) = 2f(x) f'(x) - 2\phi(x) \phi'(x)$$

$$= 2f(x) \phi(x) - 2\phi(x) f(x) = 0$$

$$h(x) = \text{constant}$$

$$f(3) = 5$$

$$f'(3) = \phi(3) = 4$$

$$h(3) = f^2(3) - \phi^2(3)$$

$$= 25 - 16 = 9$$

$$\boxed{h(x) = 9}$$

3. Ans (1.00)

$$f - f' = f' - f''$$

$$c + x = \ln(f - f')$$

$$f - f' = Ae^x \Rightarrow \frac{dy}{dx} - y = -Ae^x$$

$$y \cdot e^{-x} = \int -Ae^x \cdot e^{-x} dx$$

$$ye^{-x} = -Ax + B \quad y(0) = 0, y(1) = e$$

$$\Rightarrow y = xe^x \Rightarrow \int_0^1 xe^x dx = 1$$

4. Ans (439.00)

$$\text{Clearly, } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & 2 & p \end{vmatrix} = 1(p-6) -$$

$$1(2p-3) + 1(4-1) \text{ (Expanding along } R_1)$$

$$\Rightarrow \Delta = -p$$

Case-I: If $p \neq 0$, then system of equations has unique solution.

Case-II: If $p = 0$, put $z = k$, we get

$$x + y = 4 - k \text{ and } 2x + y = 6 - 3k$$

$\therefore$ On solving, we get

$$x = 2 - 2k, y = 2 + k$$

Now, substituting these values of x, y and z in equation $x + 2y + pz = q$, we get

$$(2 - 2k) + 2(2 + k) + p \cdot k = q$$

$$\Rightarrow 6 + 0k = q \text{ i.e., } q = 6$$

Thus for $q \neq 6$, there is no solution and for $q = 6$, there are infinite solution.

Hence, for unique solution $p \neq 0, q \in \mathbb{R}$

$$\Rightarrow L = 20 \times 21 = 420$$

for no solution we must have $p = 0, q \neq 6$

$$\Rightarrow M = 1 \times 20 = 20$$

for infinite solution $p = 0$ and $q = 6$

$$\Rightarrow N = 1 \times 1 = 1$$

$$\Rightarrow L + M - N = 420 + 20 - 1 = 440 - 1 = 439.$$

Ans.

$$\text{Alternatively: } x + y + z = 4 \quad \dots (1)$$

$$2x + y + 3z = 6 \quad \dots (2)$$

$$x + 2y + pz = q \quad \dots (3)$$

$$\text{Solving (1) and (2)} \Rightarrow x = 2 - 2z \text{ and } y = 2$$

+ z

Put in equation (3), we get

$$pz = q - 6$$

Hence, for unique solution $p \neq 0, q \in \mathbb{R}$

$$\Rightarrow L = 20 \times 21 = 420$$

for no solution we must have $p = 0, q \neq 6$

$$\Rightarrow M = 1 \times 20 = 20$$

for infinite solution $p = 0$ and $q = 6$

$$\Rightarrow N = 1 \times 1 = 1$$

$$\Rightarrow L + M - N = 420 + 20 - 1 = 439 \text{ Ans.}]$$

5. Ans (3.00)

$$f'(x) = f^2(x) \Rightarrow \frac{f'(x)}{f^2(x)} = 1 \Rightarrow \int \frac{f'(x)}{f^2(x)} dx$$

$$= \int 1 \cdot dx + C \Rightarrow -\frac{1}{f(x)} = x + C$$

$$\text{at } x = 0, f(0) = -1 \Rightarrow C = 1$$

$$\therefore f(x) = -\frac{1}{x+1}$$

$$I = \frac{\sqrt{2}}{\pi} \int_0^1 \left(\frac{1-x}{-\frac{1}{x+1}(x^4+1)} + \frac{2}{x^4+1} \right) dx$$

$$= \frac{\sqrt{2}}{\pi} \int_0^1 \left(\frac{x^2-1}{x^4+1} + \frac{2}{x^4+1} \right) dx$$

$$= \frac{\sqrt{2}}{\pi} \int_0^1 \frac{x^2+1}{x^4+1} dx$$

$$= \frac{\sqrt{2}}{\pi} \int_0^1 \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}} dx = \frac{\sqrt{2}}{\pi} \int_0^1 \frac{\left(1+\frac{1}{x^2}\right) dx}{\left(x-\frac{1}{x}\right)^2+2},$$

$$x - \frac{1}{x} = t \Rightarrow \left(1 + \frac{1}{x^2}\right) dx = dt$$

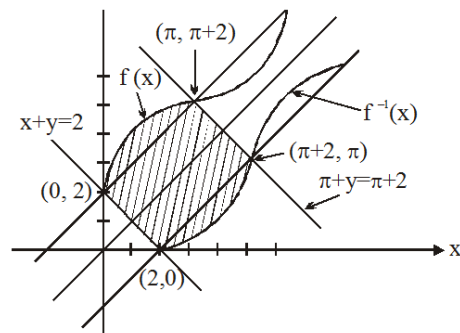
$$= \frac{\sqrt{2}}{\pi} \int_{-\infty}^0 \frac{dt}{t^2+2} = \frac{\sqrt{2}}{\pi} \frac{1}{\sqrt{2}} \left(\tan^{-1} \left(\frac{t}{\sqrt{2}} \right) \right)_{-\infty}^0$$

$$= \frac{1}{\pi} \left(0 - \left(-\frac{\pi}{2} \right) \right) = \frac{1}{2} = \frac{p}{q}$$

$$\therefore (p+q) = 3 \text{ Ans.}$$

6. Ans (4.00)

$$f(x) = \sin x + x + 2$$



$$f'(x) = \cos x + 1$$

$$f''(x) = -\sin x$$

Required Area = 2 + 2 + Area of rectangle

$$= 4 + 2\sqrt{2} \cdot \pi\sqrt{2}$$

$$= 4(\pi + 1) \equiv k(\pi + 1)$$

$$\therefore k = 4.]$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (0)

$$\text{We have, } x + y = \frac{dy}{dx} + 5 \Rightarrow \frac{dy}{dx} - y = x - 5$$

$$\text{I.F.} = e^{\int (-1)dx} = e^{-x}$$

$\therefore$ Solution

$$y(e^{-x}) = \int (x-5)e^{-x} dx$$

$$ye^{-x} = -xe^{-x} - e^{-x} + 5e^{-x} + C$$

$$\Rightarrow y = -x + 4 + Ce^x$$

Passing (0, 2)

$$\Rightarrow 2 = 4 + C \Rightarrow C = -2$$

$$\therefore y = -x + 4 - 2e^x$$

$$f'(x) = -1 - 2e^x = 0$$

$$e^x = \frac{-1}{2} \Rightarrow \text{No solution}$$

Number of real solution = 0 Ans.]

2. **Ans (5)**

$$\alpha(t) + \beta'(t) = 1 \quad \dots(1)$$

$$\alpha'(t) + \beta(t) = 1 \quad \dots(2)$$

Add (1) and (2)

$$(\alpha + \beta) + (\alpha' + \beta') = 2$$

$$\therefore (\alpha + \beta) + (\alpha + \beta)' = 2$$

$$(\alpha + \beta) = y$$

$$\therefore \frac{dy}{dt} + y = 2 \quad \text{I.F.} = e^t$$

$$y \cdot e^t = 2e^t + C$$

$$(\alpha(t) + \beta(t)) e^t = 2e^t + C$$

$$\text{Put } t = 0$$

$$2 + 1 = 2 + C$$

$$C = 1$$

$$\therefore \alpha(t) + \beta(t) = 2 + e^{-t} \quad \dots(3)$$

$$\text{|||ly (1) - (2)}$$

$$(\alpha - \beta) - (\alpha - \beta)' = 0$$

$$(\alpha - \beta)' - (\alpha - \beta) = 0$$

$$\frac{dy}{dt} - y = 0 \quad \text{I.F.} = e^{-t}$$

$$ye^{-t} = C$$

$$y = C e^t$$

$$\alpha(t) - \beta(t) = C e^t \quad \text{Put } t = 0$$

$$2 - 1 = C \quad \therefore C = 1$$

$$\alpha(t) - \beta(t) = e^t \quad \dots(4)$$

$\therefore$ Put $t = \ln 2$ in (3) & (4) and add

$$2\alpha(\ln 2) = 2 + \frac{1}{2} + 2 = 4 + \frac{1}{2} = \frac{9}{2}$$

$$\alpha(\ln 2) = \frac{9}{4} = \frac{p}{q}$$

$$\therefore p - q = 9 - 4 = 5 \text{ Ans.}]$$

3. **Ans (1)**

A is an orthogonal matrix

$$\Rightarrow a^2 + b^2 + c^2 = 1, ac + ba + cb$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 - 2(ab + bc + ca)$$

$$= 1 - 0 = 1$$

$$\Rightarrow |a + b + c| = 1$$

4. **Ans (3)**

$$XY = AB^T BA^T = I$$

$$\Rightarrow \text{tr}(XY) = 3$$

5. **Ans (3)**

$$\text{tr}(A) = \text{tr}(B)$$

$$\alpha^2 + \beta^2 + \gamma^2 = 2\alpha + 2\beta + 2\gamma - 3$$

$$(\alpha - 1)^2 + (\beta - 1)^2 + (\gamma - 1)^2 = 0$$

$$\alpha = 1 = \beta = 1 = \gamma$$

6. **Ans (4)**

W_1	W_2	W_3
A_1	•	A_3
A_2	•	A_4
	•	A_5
•		•
•		•

$$\begin{aligned} \text{Required number of ways} &= \frac{7!}{2! 2! 2! 3!} \times 2! \times \\ &2! \times 3! \times 2! \times 2! \times 3! \\ &= 12(7!)^0 k(7!) \end{aligned}$$

$$\therefore \left[\frac{k}{3} \right] = \left[\frac{12}{3} \right] = 4 \text{ Ans.}$$